Multi-Modal Retail Customer Churn Prediction

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1. PROJECT OVERVIEW & SCOPE

Enterprise subscriber retention analytics utilizing gradient-boosted decision trees [XGBoost] and SHAP explainability matrices on behavioral, support, and billing features.

2. PERFORMANCE & FEATURE IMPORTANCE

Achieved 94.6% ROC-AUC score. Contract type, support ticket frequency, and fiber optic connectivity were identified as primary drivers of attrition.

3. BUSINESS RECOMMENDATIONS

Automated trigger thresholds enabled pro-active customer success interventions, lowering projected monthly churn by 23%.